

CHASSERAL, Switzerland

WMO: 066050

Lat: 47.1317N

Lon: 7.0544E

Elev: 1631

StdP: 83.21

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.1	-11.6	-26.9	0.4	-0.7	-22.7	0.6	-1.9	38.9	-1.4	34.4	-0.9	9.3	60	0.985

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.9	21.4	14.7	19.9	14.3	18.5	13.6	15.9	19.7	15.0	18.6	14.2	17.5	5.1	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.4	12.5	17.9	13.6	11.8	16.9	12.7	11.2	16.0	50.9	19.8	48.0	18.7	45.3	17.6	20.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.0	21.8	18.4	-14.1	23.6	2.0	1.6	-15.5	24.8	-16.7	25.7	-17.8	26.7	-19.2	27.9		
			-14.9	18.0	1.9	0.9	-16.3	18.6	-17.4	19.1	-18.5	19.6	-19.9	20.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.6	-2.7	-2.8	-0.5	3.3	6.4	10.8	12.5	12.2	8.7	6.2	1.3	-1.2	
	DBStd	6.99	4.31	4.56	4.55	4.47	4.31	4.29	3.90	4.05	3.92	4.40	4.71	4.58	
	HDD10.0	2302	395	357	326	203	129	43	19	23	70	128	262	347	
	HDD18.3	5036	653	591	584	450	371	229	184	194	288	375	512	605	
	CDD10.0	317	0	0	0	3	16	66	97	92	32	11	1	0	
	CDD18.3	8	0	0	0	0	0	1	3	4	0	0	0	0	
	CDH23.3	6	0	0	0	0	0	0	4	2	0	0	0	0	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	8.7	10.6	10.4	9.3	7.8	7.9	7.3	7.1	7.0	7.3	8.8	9.8	10.9	
Precipitation	PrecAvg	1211	89	86	80	91	111	108	115	116	100	103	103	110	
	PrecMax	1569	227	205	277	179	207	220	242	274	182	184	251	237	
	PrecMin	871	21	26	22	21	50	59	50	22	39	18	11	2	
	PrecStd	177	54	45	58	49	45	43	42	52	42	43	58	55	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.2	9.6	10.5	15.3	18.6	22.4	23.7	23.3	18.3	16.3	12.9	9.2	
		MCWB	0.8	1.3	4.1	9.1	12.8	15.6	15.5	14.5	13.2	9.8	5.5	1.5	
	2%	DB	6.1	6.8	8.5	12.7	16.4	20.3	21.2	21.5	16.7	14.4	10.6	7.7	
		MCWB	0.6	0.7	3.4	7.2	11.1	14.7	14.7	14.5	12.4	8.7	4.6	0.8	
	5%	DB	4.5	4.9	6.9	11.2	14.5	18.7	19.5	19.8	15.4	13.0	9.0	6.3	
		MCWB	0.6	-0.1	2.4	6.4	9.9	13.8	14.2	14.2	11.6	8.1	4.3	0.7	
	10%	DB	3.0	3.1	5.5	9.6	12.6	17.2	18.1	18.0	14.2	11.7	7.5	4.7	
		MCWB	0.2	-0.4	1.9	5.4	8.8	12.8	13.4	13.7	11.0	7.5	3.9	0.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.5	3.4	5.5	9.5	13.6	17.1	17.0	17.0	14.8	11.4	8.0	4.7
			MCDB	5.4	5.1	7.9	13.7	17.7	20.9	21.2	20.9	16.9	13.5	10.3	5.7
2%		WB	3.1	2.1	4.5	8.0	11.8	15.6	15.7	15.7	13.4	10.3	6.5	3.5	
		MCDB	4.0	3.8	7.0	11.7	15.1	19.1	19.6	19.5	15.4	12.3	8.3	4.6	
5%		WB	2.0	1.2	3.6	6.9	10.5	14.5	14.8	14.8	12.4	9.5	5.6	2.6	
		MCDB	2.8	3.3	5.8	10.3	13.6	17.8	18.4	18.3	14.6	11.5	7.4	4.1	
10%		WB	0.8	0.3	2.6	5.9	9.2	13.3	13.9	13.9	11.4	8.6	4.6	1.8	
		MCDB	2.3	2.6	4.7	9.1	12.1	16.3	17.3	17.3	13.6	10.6	6.6	3.5	
Mean Daily Temperature Range		5% DB	MDBR	4.4	4.2	4.6	5.3	5.7	5.8	5.9	5.4	4.9	4.8	4.4	4.5
			MCDBR	5.7	5.6	5.7	6.9	7.4	6.9	7.2	6.3	6.0	5.7	5.6	5.7
	MCWBR		3.8	3.7	4.0	4.9	5.3	5.6	5.2	5.1	5.3	4.5	3.8	3.4	
	5% WB	MCDBR	4.9	5.1	5.2	6.6	7.0	6.7	6.6	6.2	5.6	5.0	4.9	4.7	
		MCWBR	4.6	4.2	4.0	4.9	5.2	5.7	5.1	5.2	5.2	4.4	4.2	4.0	
Clear-Sky Solar Irradiance	taub		0.249	0.268	0.294	0.321	0.329	0.332	0.331	0.326	0.310	0.294	0.269	0.247	
	taud		2.519	2.487	2.477	2.421	2.430	2.446	2.434	2.463	2.498	2.540	2.559	2.522	
	Ebn at Noon		874	920	938	935	935	931	926	919	908	874	839	833	
	Edh at Noon		60	77	92	107	110	109	109	101	89	73	58	54	
All-Sky Solar Radiation	RadAvg		1.19	2.02	3.22	4.35	5.03	5.95	5.74	5.00	3.83	2.43	1.32	0.96	
	RadStd		0.15	0.31	0.48	0.64	0.47	0.55	0.51	0.46	0.35	0.23	0.17	0.17	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (108 neighbors)	N/A	N/A	N/A	+0.73	+0.40	+0.30	N/A	N/A	+53	+16

Nomenclature: See separate page